# First Image of a Black Hole

The Event Horizon Telescope (EHT) — a planet-scale array of eight ground-based radio telescopes forged through international collaboration — was designed to capture images of a black hole. In coordinated press conferences across the globe, EHT researchers revealed that they succeeded, unveiling the first direct visual evidence of the supermassive black hole in the centre of Messier 87 and its shadow.

The shadow of a black hole seen here is the closest we can come to an image of the black hole itself, a completely dark object from which light cannot escape. The black hole's boundary — the event horizon from which the EHT takes its name — is around 2.5 times smaller than the shadow it casts and measures just under 40 billion km across. While this may sound large, this ring is only about 40 microarcseconds across — equivalent to measuring the length of a credit card on the surface of the Moon.

Although the telescopes making up the EHT are not physically connected, they are able to synchronize their recorded data with atomic clocks — hydrogen masers — which precisely time their observations. These observations were collected at a wavelength of 1.3 mm during a 2017 global campaign. Each telescope of the EHT produced enormous amounts of data – roughly 350 terabytes per day – which was stored on high-performance helium-filled hard drives. These data were flown to highly specialised supercomputers — known as correlators — at the Max Planck Institute for Radio Astronomy and MIT Haystack Observatory to be combined. They were then painstakingly converted into an image using novel computational tools developed by the collaboration.

**Credit:**

EHT Collaboration

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## About the Image

| | |
|---|---|
| Id: | eso1907a |
| Type: | Observation |
| Release date: | 10 April 2019, 15:07 |
| Related releases: | eso2105, eso1907 |
| Related announcements: | ann19043 |
| Size: | 7416 x 4320 px |

## About the Object

| | |
|---|---|
| Name: | M87*, Messier 87 |
| Type: | Local Universe : Galaxy : Component : Central Black Hole |
| Constellation: | Virgo |
| Category: | Quasars and Black Holes |

## Image Formats

Fullsize Original
183.4 MB checksum
Large JPEG
7.7 MB
Publication TIFF 4K
8.3 MB
Publication JPEG
403.5 KB
Screensize JPEG
39.8 KB

## Zoomable

Zoomable

## Wallpapers

1024x768
53.8 KB
1280x1024
72.6 KB
1600x1200
92.6 KB
1920x1200
98.5 KB
2048x1536
144.7 KB

## Coordinates

| | |
|---|---|
| Position (RA): | 12 30 49.42 |
| Position (Dec): | 12° 23' 28.04" |
| Field of view: | 0.00 x 0.00 arcminutes |

| | |
|---|---|
| Orientation: | North is 0.0° left of vertical |

View in WorldWide Telescope:

## Colours & filters

| Band | Wavelength | Telescope |
|---|---|---|
| Millimeter | 1.3 mm | Event Horizon Telescope |